Q1.

(a) Vinegar is sometimes put on chips. It has a sharp, sour taste.

What does the taste tell you about vinegar?

Tick the correct box.

It contains salt. ☐

It contains sugar. ☐

It is an acid. ☐

It has turned bad. ☐

1 mark

(b) Washing soda crystals react with acid to give off carbon dioxide.

If you added some washing soda crystals to vinegar, what would you **see** happening?

.....

1 mark

(c) Red cabbage can be used to test for acids and alkalis. It is bright red in acids and purple in alkalis.

(i) What colour is it in lemon juice?

.....

1 mark

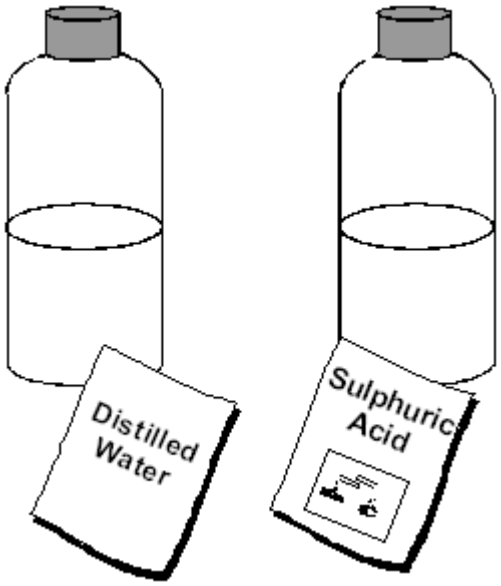
(ii) Complete the sentence below.

Substances which change colour when you add acid or alkali are

called .....

1 mark

(d) The labels have fallen off two bottles. The labels say ‘Distilled Water’ and ‘Sulphuric Acid’.



(i) Why should you **not** taste the liquids to see which is which?

.....

.....

1 mark

(ii) You can use some washing soda crystals to find out what is in each bottle. Describe what you would **do** and what you would **see** in each case.

.....

.....

.....

.....

2 marks
Maximum 7 marks

Q1.

(a) It is an acid ✓

if more than one box is ticked, award no mark

1 (L3)

(b) bubbling

*accept 'fizzing' **or** 'froth' **or** 'effervescence'*
***or** 'gas given off' **or** 'carbon dioxide given off'*
*do **not** accept 'a reaction'*

1 (L3)

(c) (i) red **or** bright red

1 (L3)

(ii) indicators

1 (L3)

(d) (i) acid can be poisonous **or** burn **or** is dangerous **or** harmful

*accept 'it makes you ill' **or** 'it is corrosive'*
***or** 'because it might be acid'*
*do **not** accept 'it will kill you'*

1 (L4)

(ii) **the answer should refer to the use of washing soda crystals with both liquids**

any **one** from

- pour some of the liquid from each bottle onto the washing soda
- add washing soda to a sample from each bottle
  *do **not** accept ' add washing soda to each bottle'*

1 (L4)

any **one** from

- one bubbles and one does not
  accept 'one of them bubbles'
- one gives off carbon dioxide and one does not
  accept 'one of them gives off carbon dioxide'
- the acid bubbles **or** gives off carbon dioxide
  *do **not** accept 'it bubbles' **or** 'it gives off carbon dioxide'*
- the water does **not** bubble **or** give off carbon dioxide

1 (L4)

[7]